QTM 220 Final Practice Problems

Problem 1

Suppose you have a sample $Y_1, Y_2 ..., Y_n$ drawn with replacement from a population with mean μ and standard deviation σ.

Consider the following estimator for μ: $\tilde{Y} = \frac{1}{n+2} \sum_{i=1}^{n} Y_i$

Part A

What is $E[\tilde{Y}]$?

$$E[\tilde{Y}] = E[\frac{1}{n+2} \sum_{i=1}^{n} Y_i]$$

$$= \frac{1}{n+2} E[\sum_{i=1}^{n} Y_i]$$

$$= \frac{1}{n+2} \sum_{i=1}^{n} E[Y_i]$$

by linearity of expectations

$$= \frac{1}{n+2} \sum_{i=1}^{n} E[Y]$$

by IID

$$= \frac{1}{n+2} \sum_{i=1}^{n} \mu$$

by definition

$$= \frac{1}{n+2} n\mu$$

$$= \frac{n}{n+2}\mu$$

Part B

How does $E[\tilde{Y}]$ compare to $E[\bar{Y}]$? (where $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$)

similar steps show that $E[\bar{Y}] = \mu$. $\bar{Y}$ is unbiased, $\tilde{Y}$ is biased, but $\lim_{n\to\infty} \frac{n}{n+2} = 1$ so bias will decrease as n increases.

Part C

Showing your work, calculate $Var[\tilde{Y}]$.

$$Var[\tilde{Y}] = V[\frac{1}{n+2}\sum_{i=1}^{n} Y_i]$$

$$= (\frac{1}{n+2})^2 V[\sum_{i=1}^{n} Y_i]$$

$$= (\frac{1}{n+2})^2 \sum_{i=1}^{n} V[Y]$$

because IID, covariances are 0

$$= (\frac{1}{n+2})^2 \sum_{i=1}^{n} \sigma^2$$

by definition

$$= (\frac{1}{n+2})^2 n\sigma^2$$

$$= \frac{n}{(n+2)^2}\sigma^2$$

Part D

How does $Var[\tilde{Y}]$ compare to $Var[\bar{Y}]$?

Similar steps show that $V[\bar{Y}] = \frac{\sigma^2}{n}$

$\frac{1}{n} > \frac{n}{(n+2)^2}$ so our $\tilde{Y}$ estimator has a lower variance than the mean. But the estimator is biased, so we probably wouldn't want to use it over the mean.

Problem 2

Suppose we have a sample of $Y_1 \dots Y_n$ and $X_1 \dots X_n$ stored in an **R** dataframe 'df' with n rows. We run this code.

```
B=10000
beta_0<- rep(NA,B)
beta_1<- rep(NA,B)
for( b in 1:B ){
bootind = sample(1:n,size=n,replace=T)
dfboot = df[bootind,]
mod<- lm(y~x, data=dfboot)
beta_0[b] = mod$coef[1]
beta_1[b] = mod$coef[2]
}
```

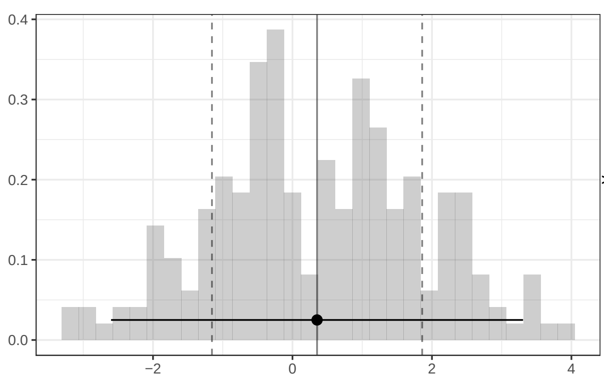

Figure 1: Histogram of the vector Y's elements

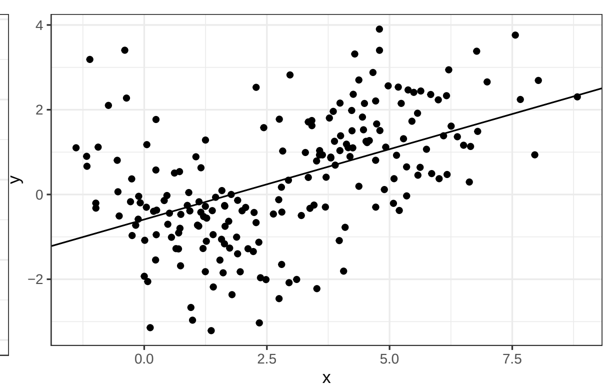

Figure 2: Scatterplot of X and Y

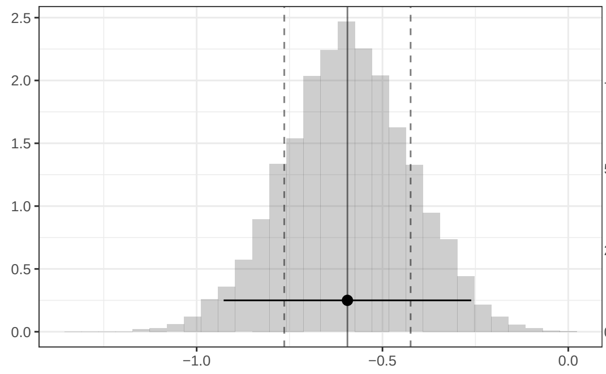

Figure 3: Histogram of the beta_0's elements

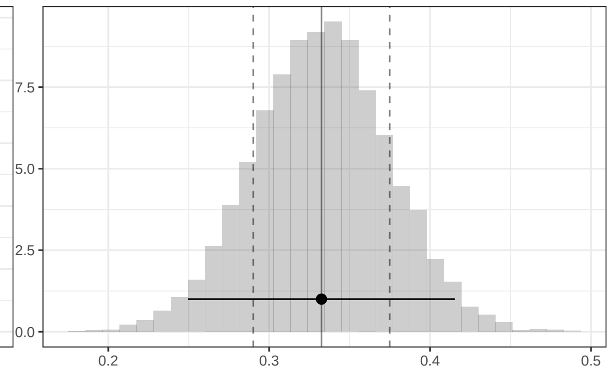

Figure 4: Histogram of the beta_1's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

The sample mean. It's indicated by the solid vertical line on the sample histogram.

Part b

What does this code calculate? State it in words and mark it with a 'b', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

The sample standard deviation of Y. It's the distance between the solid vertical and either dashed vertical line on the sample histogram.

Part c

What does this code calculate? State it in words and mark it with a 'c', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mod<- lm(y~x, data=df)
mod$coef[1]
```

The intercept of the simple regression of y on x. It is the intercept of the line on the scatterplot

Part d

What does this code calculate? State it in words and mark it with a 'd', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(beta_1)
```

The bootstrap sampling distribution's standard deviation for the coefficient on x from the simple regression (slope). Estimated standard error. It's the distance between the solid vertical and either dashed vertical line on the beta_1 sampling distribution histogram.

Part e

What does this code calculate? State it in words and mark it with a 'e', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mod$coef[2] + sqrt(diag(solve(t(X)%*%X)%*%t(X)%*%diag(resid^2)%*%X%*%solve(t(X)%*%X)))[2]

mod$coef[2] - sqrt(diag(solve(t(X)%*%X)%*%t(X)%*%diag(resid^2)%*%X%*%solve(t(X)%*%X)))[2]
```

A 95% confidence interval — based on normal approximation — for the least squares slope from the simple regression of y on x for the population from which this sample was drawn. This is indicated by the horizontal segment (with a dot in the middle) on the sampling distribution histogram (beta_1).

Part f

How could you calculate roughly the same thing as in part e using 'sd(beta_1)'

We can use the standard deviation of the bootstrap sampling distribution as our estimate of standard error.

```
mod$coef[2]  - 1.96 * sd(beta_1)
mod$coef[2]  + 1.96 * sd(beta_1)
```